

RYTHU SETHU - Bridging farmers and customers

A Mini-Project report submitted in partial fulfillment of the Academic requirements for the award of the Degree of

BACHELOR OF ENGINEERING IN INFORMATION TECHNOLOGY

By

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DEPARTMENT OF INFORMATION TECHNOLOGY

MVSR ENGINEERING COLLEGE

(An Autonomous Institution)

Nadergul, Balapur Mandal, Hyderabad-501510 2025-2026



CERTIFICATE

*This is to certify that the Mini Project-II work entitled **RYTHUSETHU - Bridging farmers and customers** is a bonafide work carried out by **Mr. K. Charan Sai Reddy (2451-23-737-001)**, **Mr. G. Harshith goud (2451-23-737-041)**, **Mr. D. Jayanth (2451-23-737-059)** in partial fulfilment of the requirements for the award of degree of Bachelor of Engineering in Information Technology from MVSR Engineering College, an Autonomous Institution, Hyderabad, during the Academic Year 2025-2026 under our guidance and supervision.*

The results embodied in this report have not been submitted to any other university or institute for the award of any degree or diploma.

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This is to certify that the work reported in the present mini project entitled **RythuSethu – Bridging Farmers and Customers** is a record of bonafide work done by us in the Department of Information Technology, MVSR Engineering College, an Autonomous Institution, affiliated to Osmania University. The reports are based on the project work done entirely by us and not copied from any other source. The results embodied in this project report have not been submitted to any other University or Institute for the award of any degree or diploma to the best of our knowledge and belief.

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Finally, we would like to take this opportunity to thank our family for their support through the work. We sincerely acknowledge and thank all those who gave directly or indirectly their support in the completion of this work.

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VISION & MISSION, PROGRAM EDUCATIONAL OUTCOMES

Vision of the Department:

To impart technical education producing competent and socially responsible engineering professionals in the field of Information Technology.

Mission of the Department:

M1. To make teaching learning process effective and stimulating.

M2. To provide adequate fundamental knowledge of sciences and Information Technology with positive attitude.

M3. To create an environment that enhances skills and technologies required for industry. M4. To encourage creativity and innovation for solving real world problems.

M5. To cultivate professional ethics in students and inculcate a sense of responsibility towards society
Program Educational Objectives:

After 3 to 4 years of graduation, graduates of the Information Technology program will:

- I. Apply knowledge of mathematics and Information Technology to analyze, design and implement solutions for real world problems in core or in multidisciplinary areas.
- II. Communicate effectively, work in a team, practice professional ethics and apply knowledge of computing technologies for societal development.
- III. Engage in Professional development or postgraduate education to be a life-long learner.

PROGRAM OUTCOMES (POs)

PROGRAM SPECIFIC OUTCOMES (PSOs):

PSO1: **Hardware design:** An ability to analyse, design, simulate and implement computer hardware / software and use basic analog/digital circuits, VLSI design for various computing and communication system applications.

PSO2: **Software design:** An ability to analyze a problem, design algorithm, identify and define the computing requirements appropriate to its solution and implement the same

COURSE OBJECTIVES & COURSE OUTCOMES

COURSE OBJECTIVES

- Enable students learning by doing.
- Develop capability to analyse and solve real time problems.
- Inculcate innovative ideas.
- Impart team building and management skills.
- Instil writing and presentation skills.

COURSE OUTCOMES

After completion of the course student will be able to:

CO1-Interpret Literature with the purpose to formulate a problem

statement. CO2 - Plan, Analyse, Design and implement a project.

CO3-Find the solution of identified problem with the help of modern tools & technology. CO4 - Plan to work as a team to exhibit team skills to meet deadlines.

CO5 - Prepare and submit the Report and deliver presentation.

ABSTRACT

Agriculture is a vital sector of the Indian economy, but traditional supply chains and the involvement of intermediaries often reduce farmers' profits and increase costs for consumers. In addition, the lack of data-driven decision-making makes farmers vulnerable to market fluctuations, improper crop selection, and unpredictable conditions. **Rythu Sethu** is an AI-driven, multilingual agricultural e-commerce platform that directly connects farmers and consumers through a unified digital ecosystem. Developed using the **MERN stack** and **Python-based Machine Learning**, the system provides features such as voice-assisted crop listing, price and demand prediction, crop recommendation, real-time delivery tracking, and AI-powered analytics. It supports four user roles—**Farmer, Customer, Delivery Agent, and Admin**—and enables farmers to interact in their native language. By integrating **Artificial Intelligence, Machine Learning, and real-time communication**, Rythu Sethu creates a smart, transparent, and efficient farm-to-consumer marketplace that improves farmer profitability and promotes sustainable agriculture.

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Table 5.1 System Test Cases

LIST OF ABBREVIATIONS

AI - Artificial Intelligence

API - Application Programming Interface

CNN - Convolutional Neural Network

ETA - Estimated Time of Arrival

GPS - Global Positioning System

IoT - Internet of Things

JWT - JSON Web Token

LLM - Large Language Model

MERN - MongoDB, Express.js, React.js and Node.js

NLP - Natural Language Processing

ODM - Object Document Mapper

REST - Representational State Transfer

CHAPTER 1

INTRODUCTION

The agricultural sector is highly unpredictable, and farmers often rely on traditional experience rather than data-driven decisions for crop cultivation, harvest timing, and pricing. This frequently leads to market oversupply, price drops, and wastage of produce. **Rythu Sethu** was developed to address this issue by digitizing the agricultural supply chain and directly connecting farmers with consumers. The platform provides farmers with real-time insights into market demand, crop scarcity, and price trends before cultivation. Built using the **MERN stack and Python-based Machine Learning**, Rythu Sethu also includes multilingual voice assistance to overcome literacy barriers. The system creates a transparent, efficient, and intelligent agricultural ecosystem that benefits farmers, consumers, delivery agents, and administrators.

1.1 PROBLEM STATEMENT

To Develop a digital platform that directly connects farmers and customers to enable fair farmer earnings while providing customers with affordable organic food without intermediaries.

1.2 EXISTING SYSTEM

The existing agricultural platforms primarily function as basic e-commerce systems that offer limited features such as product listing and purchasing. These systems lack intelligent capabilities such as price prediction, demand forecasting, crop recommendations, and data-driven decision-making, forcing farmers to rely on traditional methods. Most platforms also depend on English-based interfaces, creating a digital literacy barrier for rural farmers. In addition, they provide limited logistics support and do not offer real-time tracking or integrated Machine Learning services. As a result, existing systems fail to provide a unified, intelligent, and farmer-friendly ecosystem that efficiently connects farmers, consumers, delivery agents, and administrators.

1.3 PROPOSED SYSTEM

Rythu Sethu is a multilingual, AI-driven agricultural e-commerce platform designed to directly connect farmers and consumers through a unified digital ecosystem. The system eliminates the dependency on intermediaries and enables efficient management of agricultural activities through a single platform. It provides intelligent support for crop listing, market analysis, price prediction, and delivery management while allowing farmers to interact using their native language. By integrating Artificial Intelligence, Machine Learning, and real-time communication technologies, the proposed system creates a smart, transparent, and data-driven farm-to-consumer marketplace that improves decision-making and enhances agricultural productivity.

1.4 SCOPE

The scope of **Rythu Sethu** encompasses the development and deployment of a full-stack, multi-tiered agricultural platform that directly connects farmers and consumers. It includes the development of a responsive Progressive Web Application (PWA) using React and Vite, along with a secure Node.js and Express backend for handling REST API requests and user authentication. The project also covers the integration of Machine Learning models for price prediction, demand forecasting, and crop recommendation, as well as a centralized Admin dashboard for model retraining and system monitoring. In addition, it supports multilingual voice assistance and real-time delivery tracking through WebSocket communication, creating a smart, efficient, and data-driven farm-to-consumer ecosystem.

CHAPTER 2

SYSTEM REQUIREMENT SPECIFICATIONS

2.1 SOFTWARE REQUIREMENTS

- **Operating System: Windows 10/11.**
- **Frontend Framework: React.js 18 with Vite.**
- **Backend Framework: Node.js and Express.js.**
- **Programming Language: Python 3.10+.**
- **Database Management System: MongoDB with Mongoose ODM. Machine Learning Libraries: Scikit-Learn and PyMongo.**
- **Real-Time Communication: Socket.io. Authentication: JSON Web Token (JWT).**
- **Voice Assistance: Web Speech API. Development Environment: Visual Studio Code.**

2.2 HARDWARE REQUIREMENTS

- **Processor: Intel Core i5 / AMD Ryzen 5 or higher.**
- **RAM: 8 GB Minimum (16 GB Recommended). Storage: 20 GB Free SSD Space.**
- **Network: Broadband Internet / Wi-Fi Connection.**
- **Client Device: Smartphone, Laptop, or Desktop Computer.**

2.3 SYSTEM ARCHITECTURE

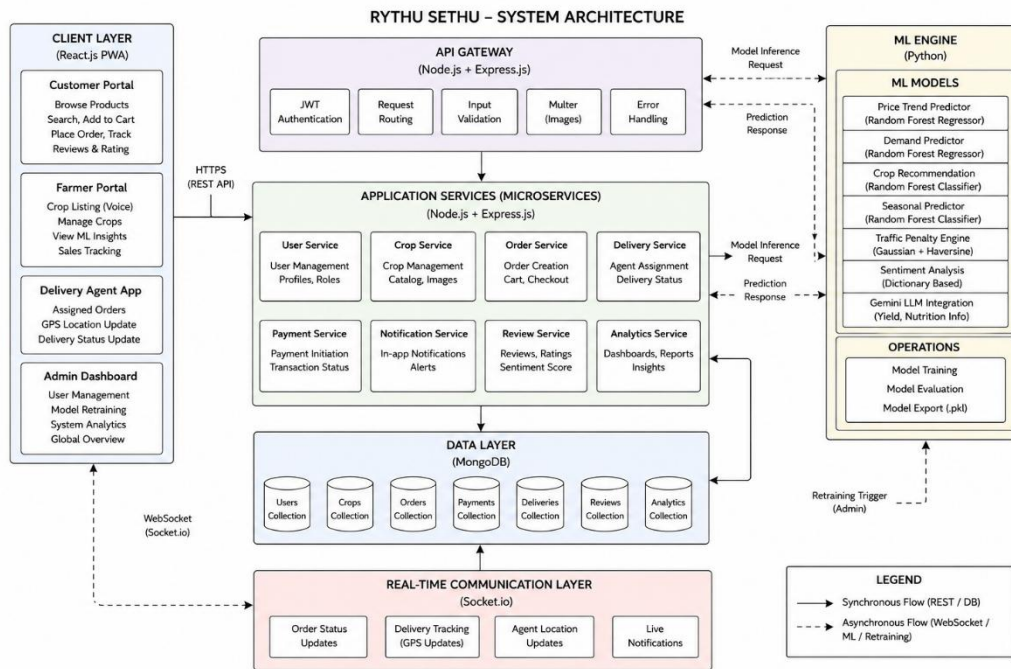


Fig-2.3.1 System Architecture

2.4 LITERATURE SURVEY

S.No	Author	Year	Title	Modules Covered	Key Points	Adopted Features	Limitations
1	I. Rahman, Riyazulla	2024	Farm to Fork	Farmer, Customer, Market	Digital farm marketplace	Farmer-consumer platform	No AI support
2	Univ. of Washington	2020	Farm-to-Table System	Farmer, Customer, Admin, Delivery	Organized supply chain	Delivery tracking	Region specific
3	S. G. Yogesh, D. S. Ravindran	2023	Organic Sales Profitability	Farmer, Customer, Admin	Improves farmer income	Direct digital sales	Literacy issues
4	S. Pudumalar et al.	2017	Crop Recommendation	Farmer, Recommendation	Soil-based crop selection	ML recommendation	Limited dataset
5	N. H. Kulkarni et al.	2018	Crop Productivity	Farmer, Analytics	Improves productivity	Ensemble learning	No real-time support
6	Y. J. N. Kumar et al.	2020	Crop Yield Prediction	Farmer, ML Engine	Predicts crop yield	Yield forecasting	Data dependent
7	N. Shirisha et al.	2024	Multilingual Farming	Farmer, Voice Module	Supports native languages	NLP assistance	Translation errors
8	S. Sultana et al.	2025	Organic Farming Landscape	Farmer, Market	Studies market trends	Buyer insights	Limited implementation

Table-2.4.1 Literature survey

CHAPTER 3

DESIGN AND IMPLEMENTATION

3.1 FEATURES

3.1.1 Price Trend Prediction

The system uses the Random Forest Regressor algorithm to predict crop prices based on market demand, supply, and seasonal factors. This helps farmers make informed decisions and reduce losses caused by market fluctuations. Algorithm Used: Random Forest Regressor

3.1.2 Crop Recommendation System

The system recommends suitable crops by analyzing soil nutrients, temperature, humidity, rainfall, and soil pH values. It helps farmers select crops that are most appropriate for their agricultural conditions. Algorithm Used: Random Forest Classifier

3.1.3 Route Optimization and ETA Prediction

The system calculates the shortest delivery route and estimates delivery time using geographical distance and traffic conditions.

Techniques Used: Haversine Formula and Gaussian Traffic Penalty Algorithm

3.1.4 Sentiment Analysis

Customer reviews are analyzed to determine positive, negative, or harmful feedback. This helps maintain platform reliability and improve trust between farmers and consumers.

Technique Used: Natural Language Processing (NLP)

3.1.5 Delivery Agent Assignment

The system automatically assigns the nearest available delivery agent by calculating the geographical distance between farms and active agents, improving delivery efficiency.

Technique Used: Haversine Distance Matching

3.1.6 Voice Assistance and Multilingual Support

The platform enables farmers to interact using their native language through voice commands. Speech is converted into text and automatically populated into the system.

Technologies Used: Web Speech API and React Context API

3.1.7 Generative AI Integration: The system integrates Google Gemini to provide intelligent agricultural assistance such as yield prediction, nutritional information, and AI-based insights.

Technology Used: Google Gemini 1.5 Flash

3.1.8 Real-Time Delivery Tracking : The platform provides live delivery updates by continuously tracking the delivery agent's location and updating customers in real time.

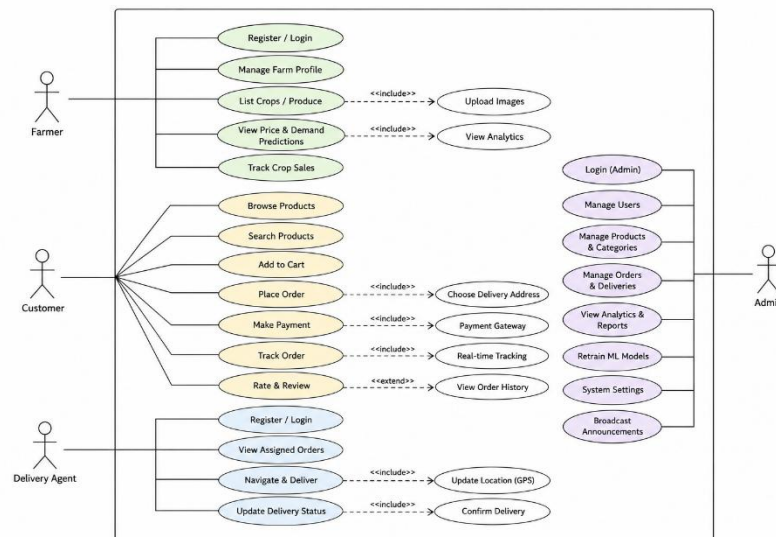
Technology Used: Socket.io

3.1.9 Image Validation and Search Suggestions: The system validates uploaded crop images and provides smart search suggestions to help customers quickly find relevant products. Techniques Used: Image Integrity Validation and AI-based Search Filtering

UML DIAGRAMS

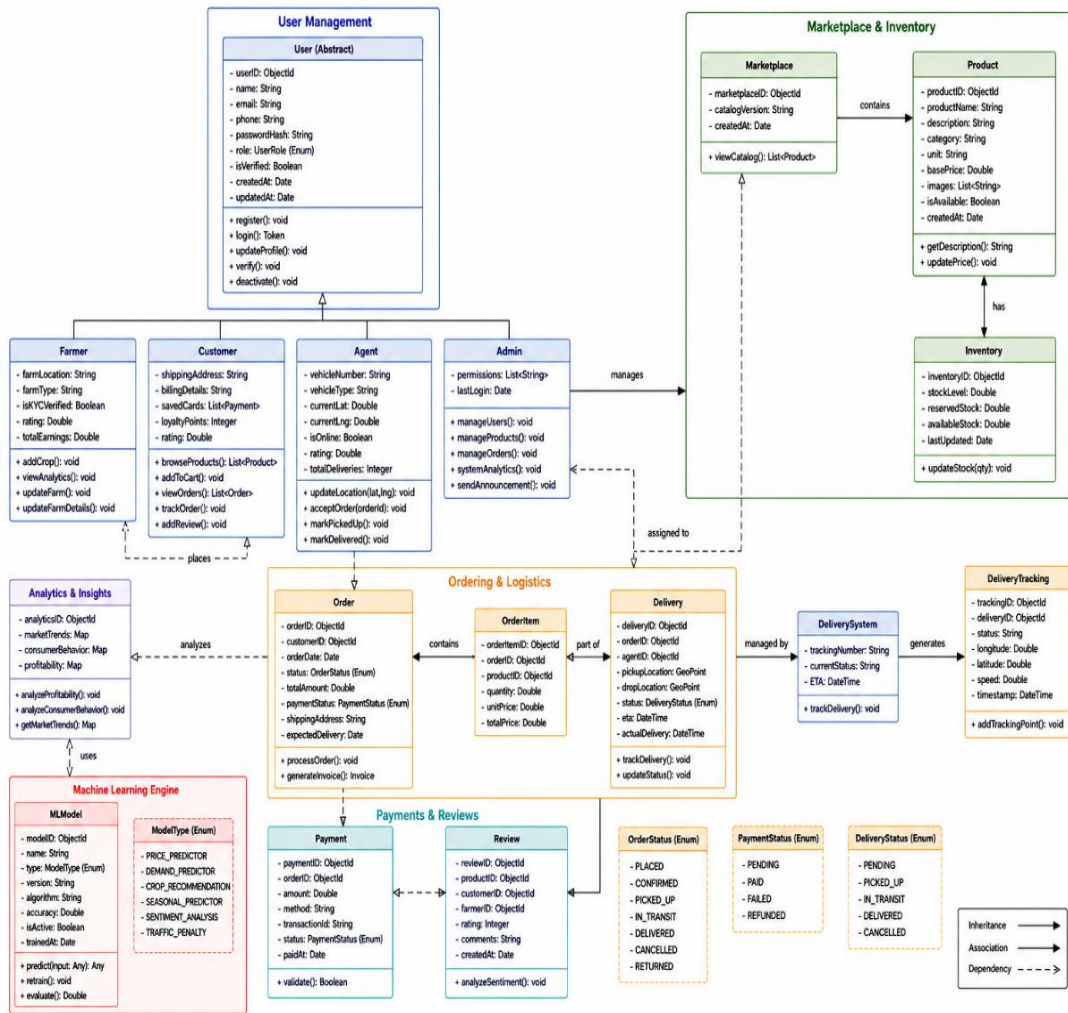
3.1.1 Use case Diagram

Fig-3.1.1



3.1.2 Class Diagram

Fig-3.1.2



3.1.3 Sequence Diagram

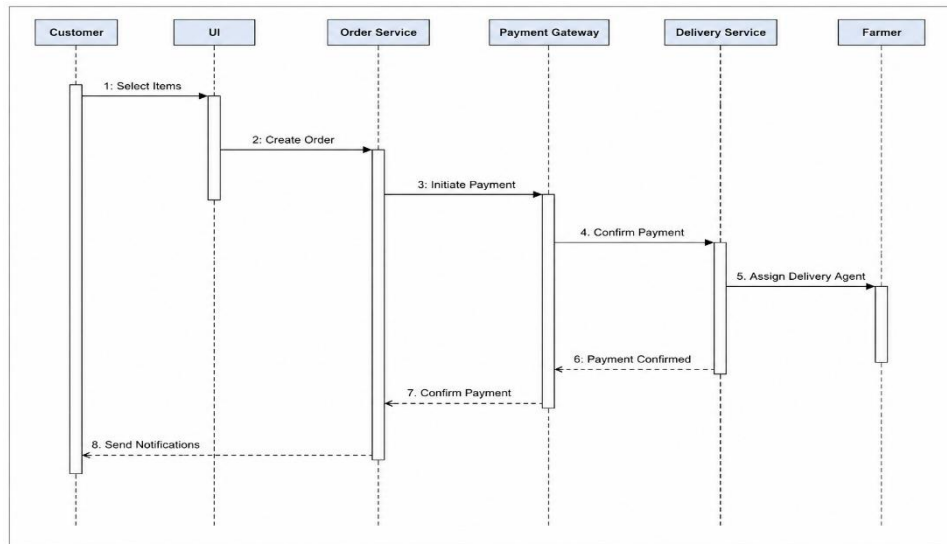


Fig-3.1.3

3.1.4 Activity Diagram

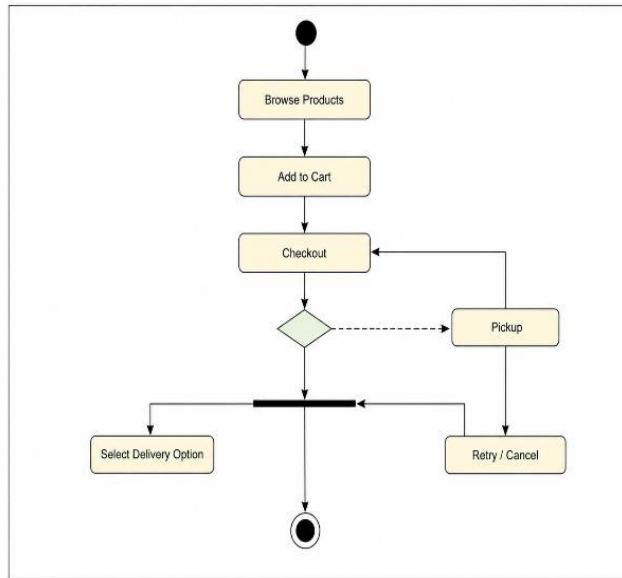


Fig-3.1.4

3.2 ENVIRONMENTAL SETUP

Installing An Editor: Visual Studio (VS) Code

1. Visit the official website of the Visual Studio Code using any web browser like Google Chrome, Microsoft Edge, etc.

Press the “**Download for Windows**” button on the website to start the download of the Visual Studio Code Application.

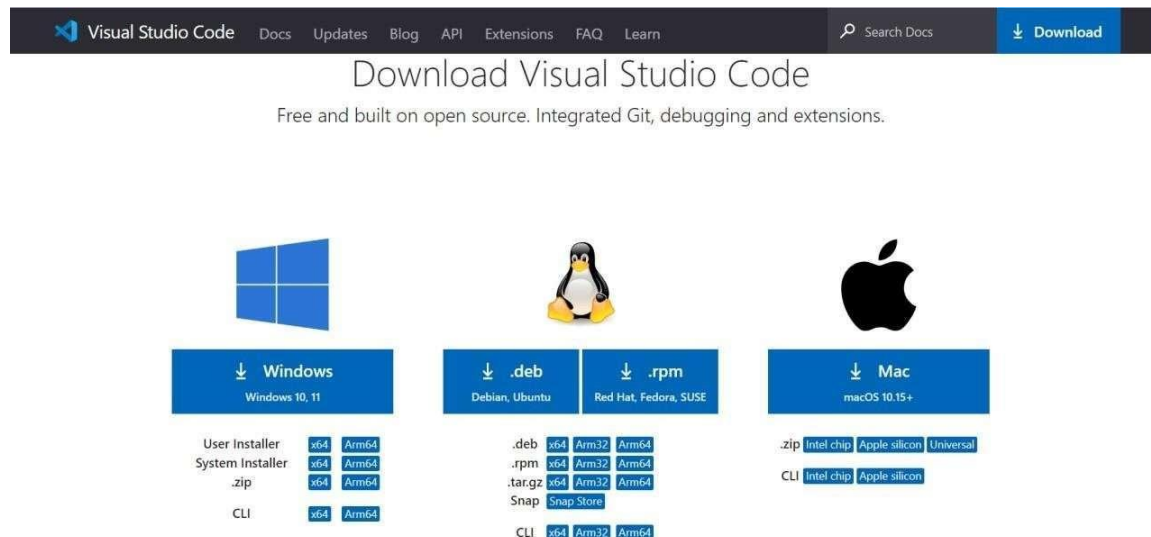


Fig-3.2.1 Visual Code Installation

2. When the download finishes, then the Visual Studio Code icon appears in the downloads folder.

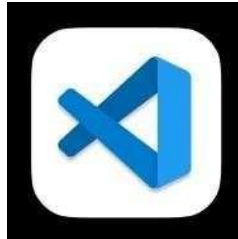


Fig-3.2.2 Vs Code Icon

3. Click on the installer icon to start the installation process of the Visual Studio Code. After the Installer opens, it will ask you for accepting the terms and conditions of the Visual Studio Code. Click on **I accept the agreement** and then click the **Next** button.

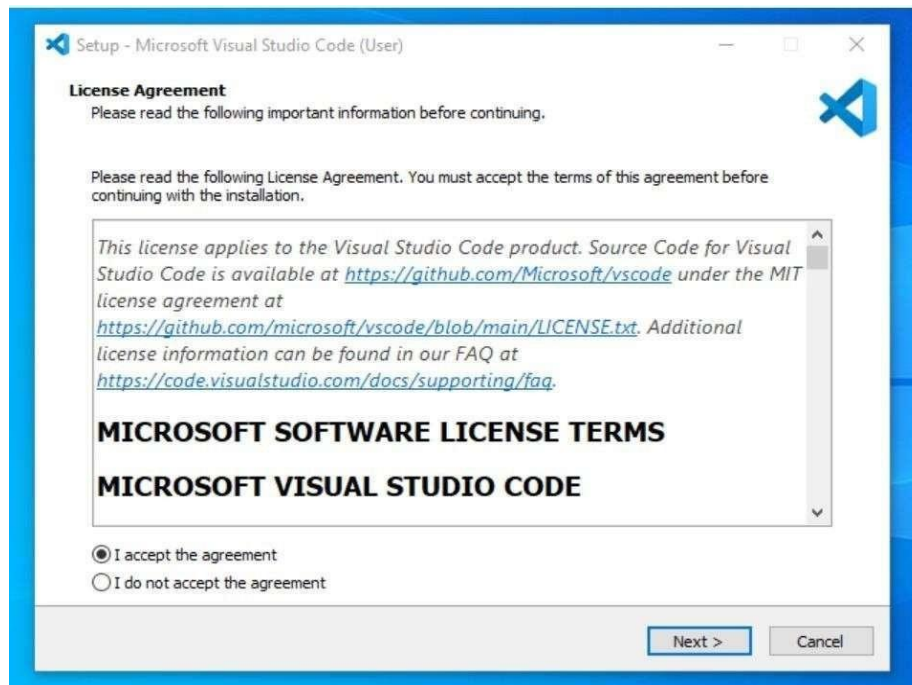


Fig-3.2.3 License Agreement

4. Choose the location data for running the Visual Studio Code. It will then ask you for browsing the location. Then click on **Next** button.

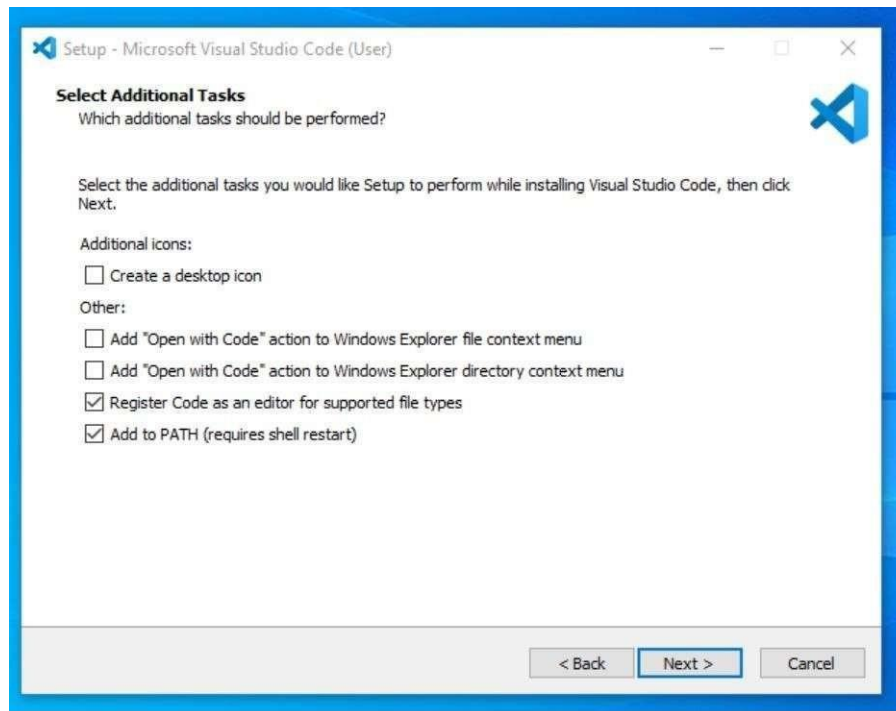


Fig-3.2.4 Additional Task

5. Then it will ask for beginning the installing setup. Click on the **Install** button.

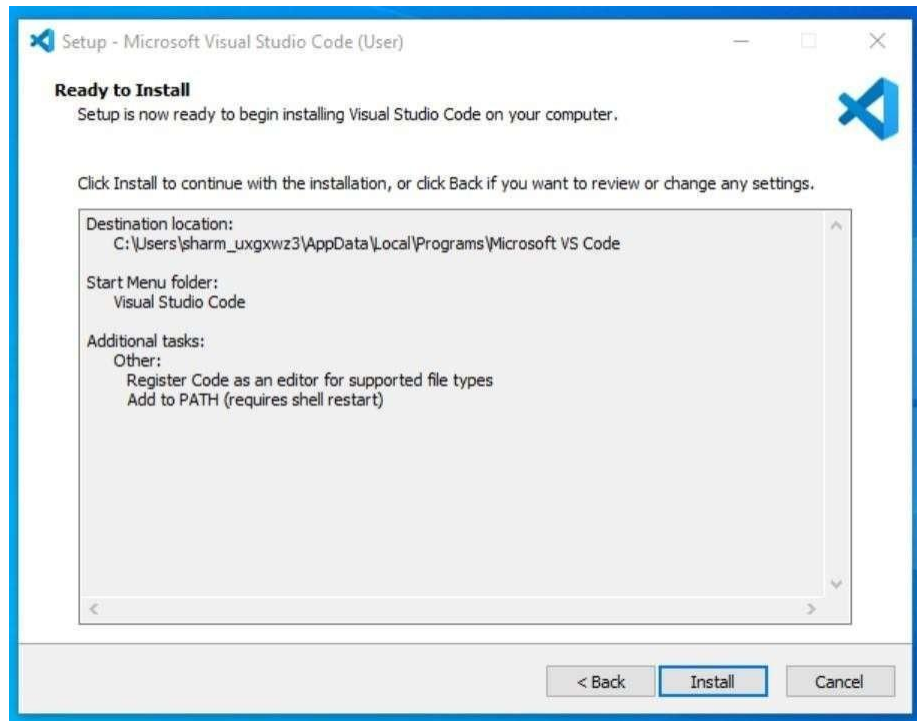


Fig-3.2.5 App Installation

6. After clicking on Install, it will take about 1 minute to install the Visual Studio Code on your device.

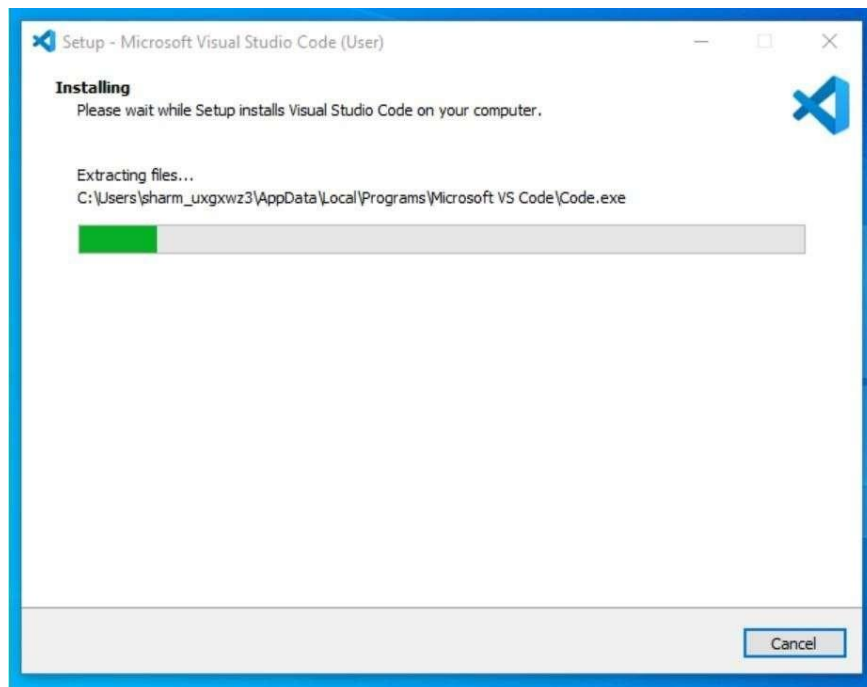


Fig-3.2.6 Vs Code Setup

7. After the Installation setup for Visual Studio Code is finished, it will show a window like this below. Tick the “**Launch Visual Studio Code**” checkbox and then click **Next**.

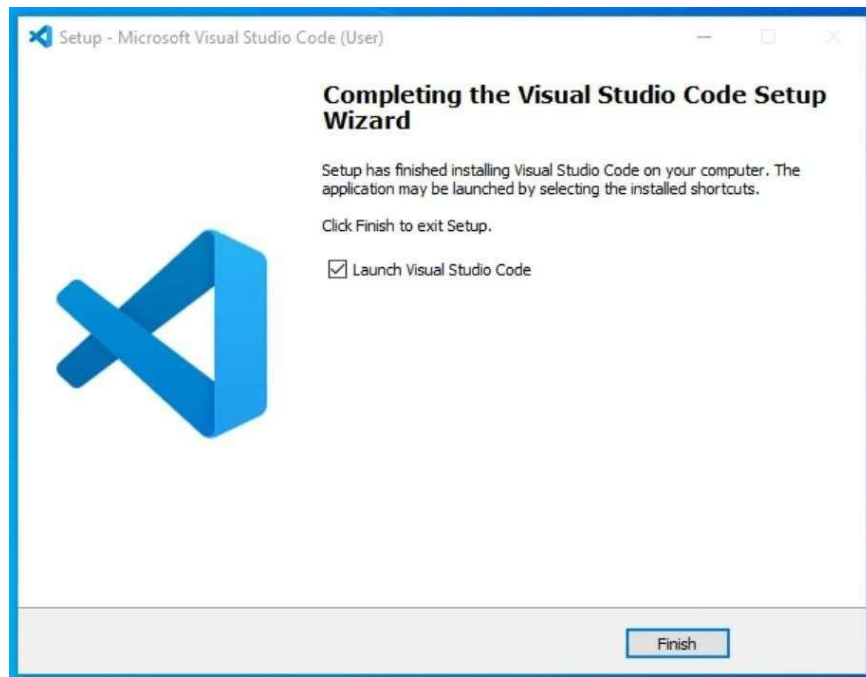


Fig-3.2.7 Vs Code Setup Wizard

8. After the previous step, the **Visual Studio Code window** opens successfully. Now you can create a new file in the Visual Studio Code window and choose a language of yours to begin your programming journey!

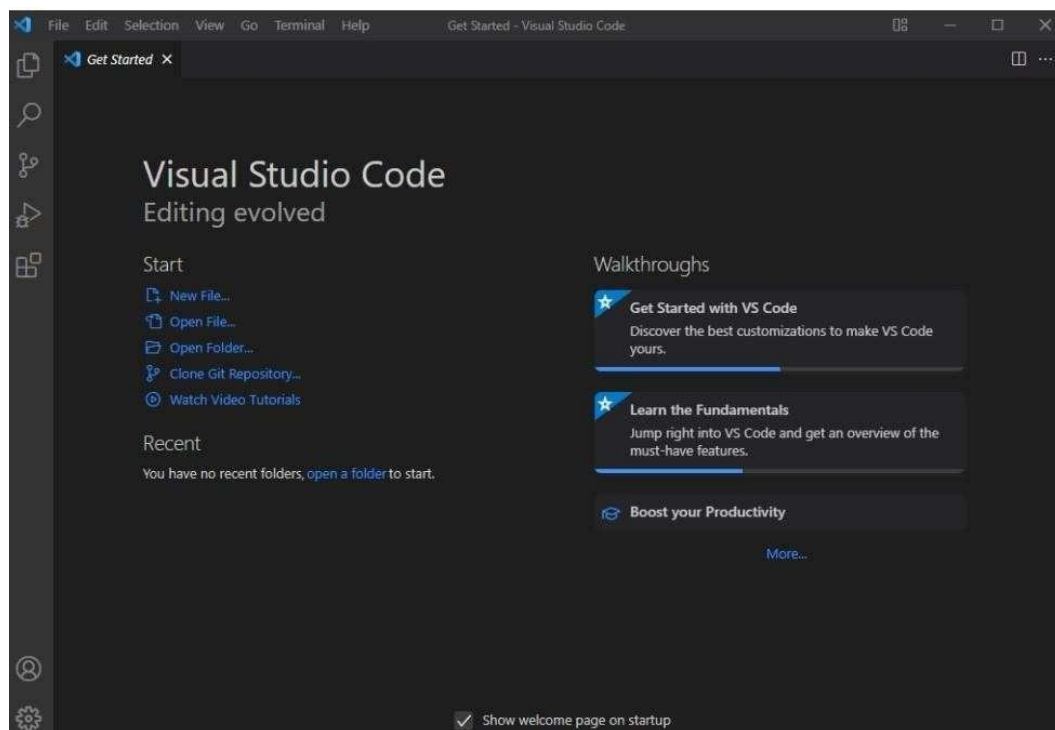


Fig-3.2.8 Vs Code Welcome Page

So this is how we successfully installed **Visual Studio Code** on our Windows system.

3.3 IMPLEMENTATION AND METHODOLOGY STEPS

The development by **Agile Software Development Life Cycle (SDLC)** to ensure a systematic, scalable, and iterative development process. Initially, the requirements and challenges faced by farmers and consumers were analyzed to define the project objectives. A multi-tier architecture was then designed to integrate the frontend, backend, database, and Machine Learning modules into a unified platform. The system was developed using the MERN stack along with Python-based Machine Learning models to provide intelligent agricultural services. Real-time communication and multilingual voice assistance were incorporated to improve accessibility and user interaction. Finally, the application was tested and validated to ensure reliability, security, and efficient performance.

- **Requirement Analysis**

The requirements of farmers, customers, delivery agents, and administrators were analyzed, and the limitations of existing agricultural systems were identified.

- **System Design**

The overall system architecture, database schema, user workflows, and UML diagrams were designed to establish a structured framework.

- **Frontend Development**

A responsive Progressive Web Application (PWA) was developed using React.js and Vite to provide an interactive and user-friendly interface.

- **Backend Development**

Node.js and Express.js were used to develop REST APIs for authentication, order management, business logic, and user management.

- **Database Integration**

MongoDB was integrated to store and manage user, crop, order, delivery, and review data.

- **Machine Learning Integration**

Python-based Machine Learning models were implemented for price prediction, demand forecasting, and crop recommendation.

- **Real-Time Communication Integration**

Socket.io was integrated to provide live delivery tracking and real-time order updates.

- **Voice Assistance Integration**

Multilingual voice support was implemented using the Web Speech API to enable farmers to interact in their native language.

- **Testing and Deployment**

The system was tested for functionality, performance, security, and accuracy before deployment.

- **Version Control:** Used Git/GitHub for code management with commands: `git add .`, `git commit`, `git push`.

CHAPTER 4

RESULTS & DISCUSSION

4.1 UI Screenshots

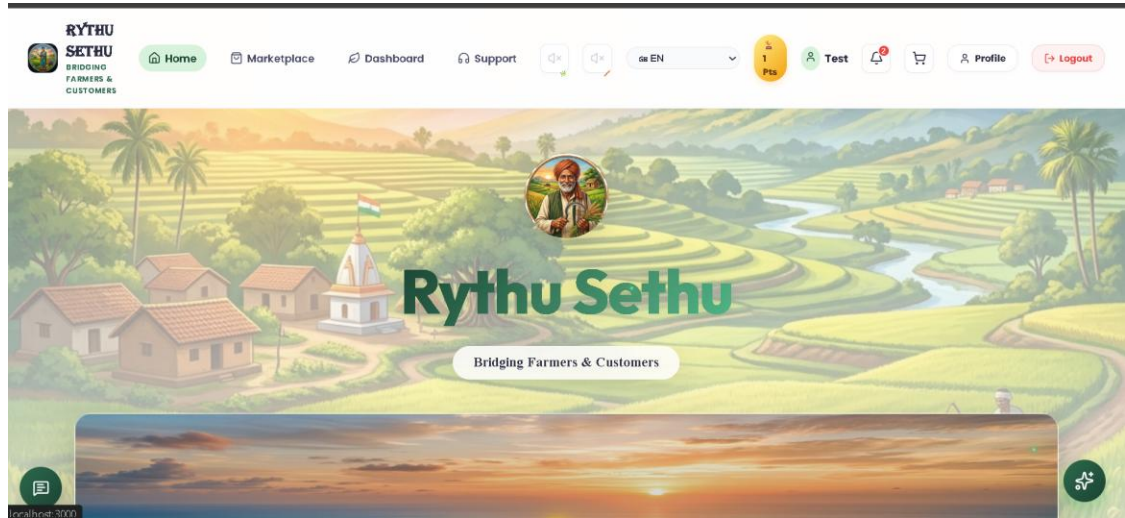


Fig-4.1 Home page

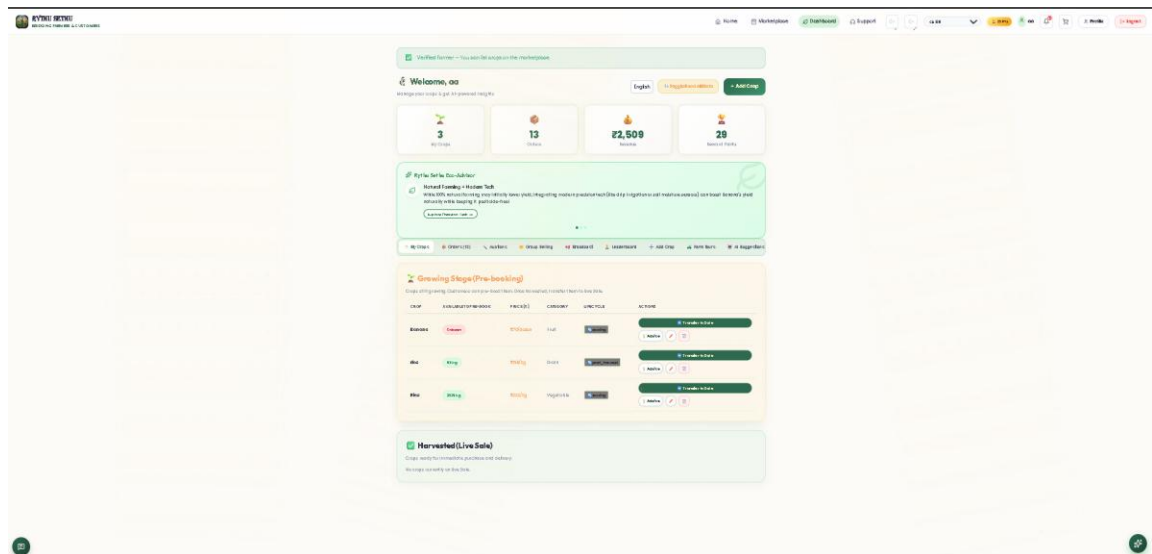


Fig-4.2 Farmer Page

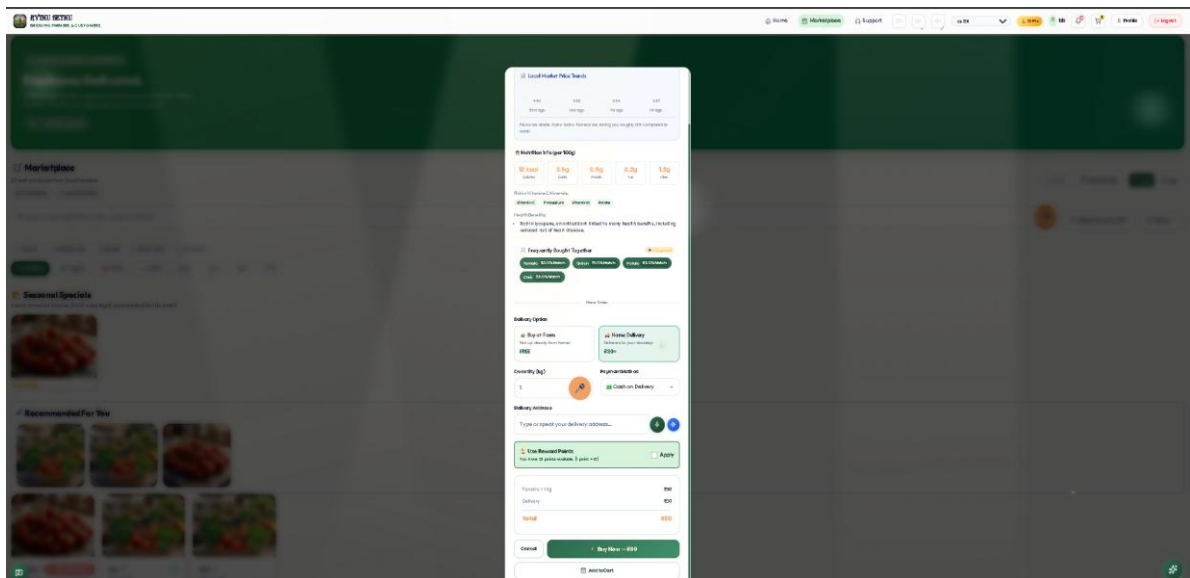


Fig-4.3 Customer page

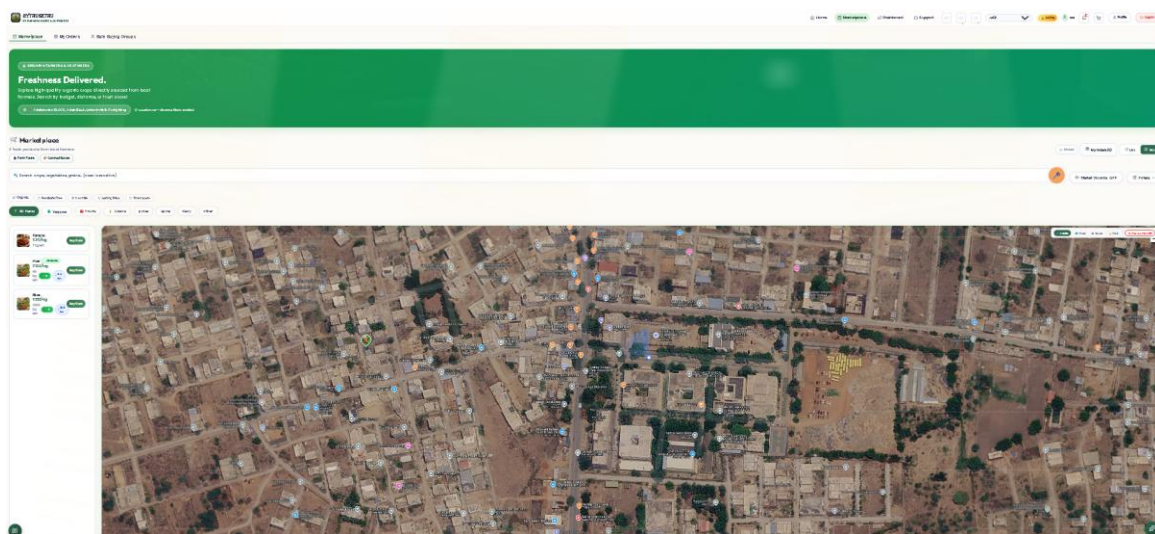


Fig-4.4 Marketplace Page

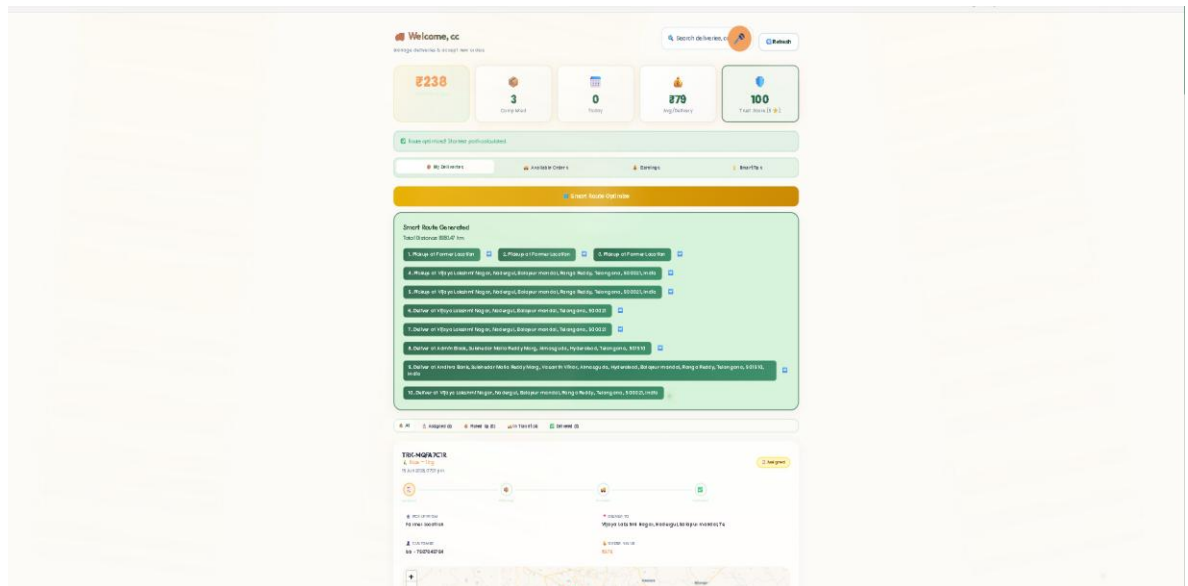


Fig-4.5 Agent dashboard

Fig-4.6 Admin Page

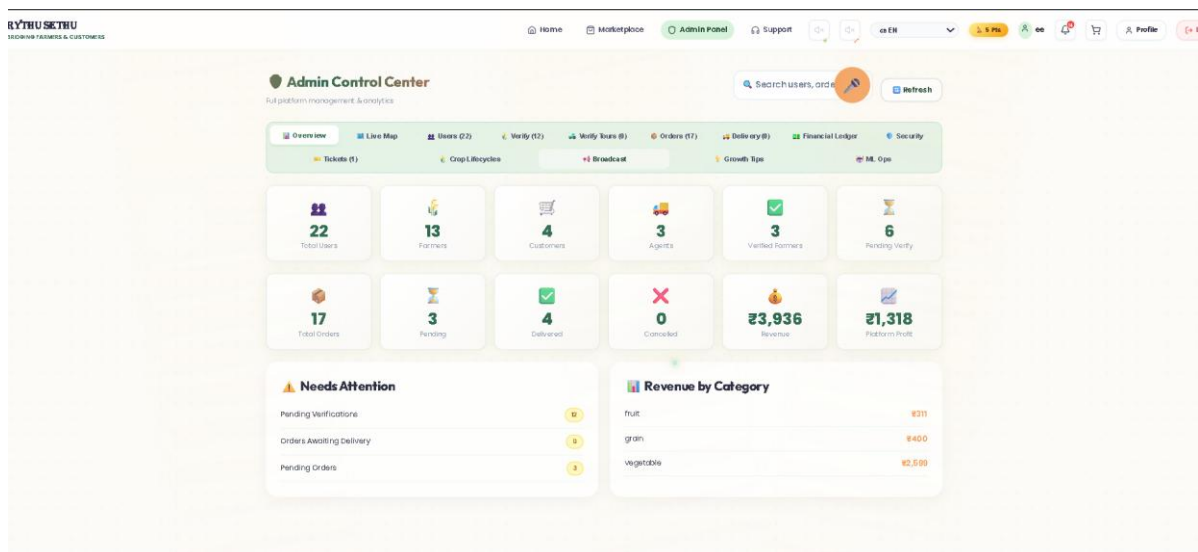


Table 5.1: System Test Cases

Test Case No.	Test Case Name	Test Case Description	Test Case Input	Test Case Output	Status
1	Database Connection	Verifies successful connection between the application and MongoDB database.	Execute <code>mongoose.connect()</code> with MongoDB running on port 27017.	Displays "MongoDB Connected" in the console.	Pass
2	Voice Authentication	Verifies voice-based user registration using speech recognition.	Grant microphone permission and say "Register as Farmer" .	Speech is recognized and the registration form is auto-filled.	Pass
3	Frontend PWA	Verifies responsive data tables on mobile devices.	Open complex data tables in mobile view.	Horizontal scrolling works without breaking the layout.	Pass
4	Real-Time Location Tracking	Verifies live location updates using Socket.io.	Trigger <code>agent_location_update</code> event from the agent application.	Consumer map updates instantly and marker moves in real time.	Pass
5	API Security (JWT Validation)	Verifies access control for protected routes.	Send a request using an invalid JWT token.	Returns 401 Unauthorized response.	Pass
6	User Login	Verifies user authentication for all roles.	Enter valid email and password credentials.	User successfully logs into the dashboard.	Pass
7	Crop Listing	Verifies crop upload by farmers.	Farmer enters crop details and uploads an image.	Crop is successfully added to the marketplace.	Pass
8	Price Prediction	Verifies AI-based crop price prediction.	Provide crop and market details.	Predicted price is displayed correctly.	Pass
9	Crop Recommendation	Verifies crop suggestion based on soil parameters.	Enter N, P, K, temperature, humidity, rainfall, and pH values.	Suitable crop recommendations are displayed.	Pass
10	Order Placement	Verifies customer order creation.	Customer adds products to the cart and places an order.	Order is successfully generated.	Pass
11	Payment Processing	Verifies payment module functionality.	Complete the payment process.	Payment is processed and order status is updated.	Pass
12	Admin Dashboard	Verifies administrative monitoring functions.	Admin accesses analytics and system controls.	Dashboard displays user, crop, and order information.	Pass

CHAPTER 5

CONCLUSION AND FUTURE ENHANCEMENTS

4.1 CONCLUSION

Rythu Sethu is an AI-driven agricultural platform that directly connects farmers and consumers through a smart digital ecosystem. The system addresses the limitations of traditional agricultural supply chains by reducing dependency on intermediaries and enabling data-driven decision-making. Features such as price prediction, crop recommendation, multilingual voice assistance, and real-time delivery tracking improve accessibility and operational efficiency. By integrating the MERN stack, Machine Learning, and real-time communication technologies, Rythu Sethu provides a transparent, efficient, and sustainable farm-to-consumer marketplace that enhances farmer profitability and promotes modern agricultural practices.

FUTURE ENHANCEMENTS

Rythu Sethu to improve its functionality and scalability in the future. **IoT-based sensors** can be integrated to automatically collect soil moisture, temperature, and pH data for accurate crop recommendations. **Blockchain technology** can be implemented to provide secure and transparent agricultural transactions. **Computer Vision techniques** can be introduced to detect plant diseases using crop images and provide instant treatment suggestions. In addition, advanced AI models can be integrated to improve demand forecasting and price prediction accuracy. The platform can also be expanded to support regional languages, government scheme integration, and nationwide deployment to create a more intelligent and sustainable agricultural ecosystem.

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APPENDIX

A: Folder structure

RythuSethu/

```
|— backend
| |— config
| | |— db.js
| | |— jwt.js
| | |— multer.js
| |— controllers
| | |— adminController.js
| | |— agentController.js
| | |— aiController.js
| | |— authController.js
| | |— cropController.js
| | |— customerController.js
| | |— deliveryController.js
| | |— farmerController.js
| | |— mlController.js
| | |— orderController.js
| | |— paymentController.js
| |— data
| | |— seedData.js
| |— middleware
| | |— authMiddleware.js
| | |— errorMiddleware.js
| | |— roleMiddleware.js
| | |— upload.js
| | |— uploadMiddleware.js
| |— models
| | |— Admin.js
| | |— Agent.js
| | |— AILog.js
```

- | | |— Auction.js
- | | |— Block.js
- | | |— BoxSubscription.js
- | | |— Cart.js
- | | |— Crop.js
- | | |— CropRequest.js
- | | |— Customer.js
- | | |— Delivery.js
- | | |— Demand.js
- | | |— Farm.js
- | | |— Farmer.js
- | | |— FarmTourBooking.js
- | | |— GlobalConfig.js
- | | |— Group.js
- | | |— Notification.js
- | | |— Order.js
- | | |— Payment.js
- | | |— Policy.js
- | | |— Prediction.js
- | | |— Review.js
- | | |— Subscription.js
- | | |— Ticket.js
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 - | | └─ deliveryRoutes.js
 - | | └─ ecommerceRoutes.js
 - | | └─ farmerRoutes.js
 - | | └─ farmTourRoutes.js
 - | | └─ groupRoutes.js
 - | | └─ mlRoutes.js

```
| | └─ notificationRoutes.js
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| | └─ policyRoutes.js
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| | └─ reportRoutes.js
| | └─ subscriptionRoutes.js
| | └─ ticketRoutes.js
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| | └─ geminiService.js
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| | └─ trustScoreService.js
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| └─ server.js
```

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 - | └─ 3_Class_Diagram_UML.png
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 - | └─ 4_Sequence_Diagram_UML.png
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└─ Project_Report.md
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└─ temp_tree.txt
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